

From a page to the vector index

Index construction

C0 uses complete non-empty pages as chunks, with whitespace normalisation and no overlap. Azure OpenAI supplies text-embedding-3-small embeddings [17].

The figures use a fictional planetarium guideline. Passages, IDs, model responses, scores, counts and grades are illustrative, not measured study results. N denotes the number of indexed chunks.

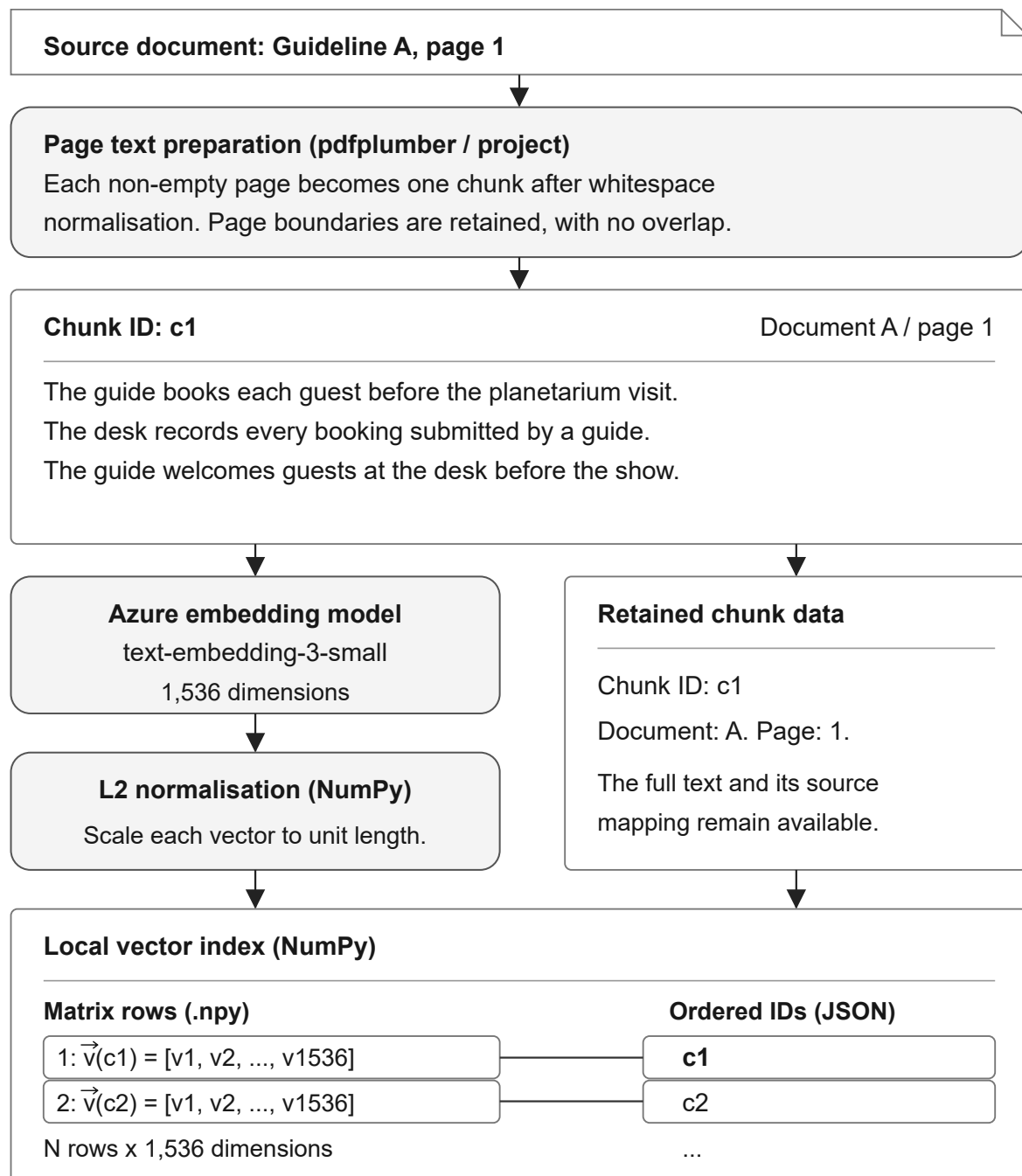


Figure 18. Ordered IDs bind matrix rows to full chunk texts and document-page mappings. Each vector has 1,536 dimensions and is normalised to unit length.

Thesis: Section 2.5; Appendix H.

Match the question to every chunk

Retrieval

The query uses the same Azure embedding model as the index [17]. After L2 normalisation, the exact dot product equals cosine similarity [38, Section 6.3].

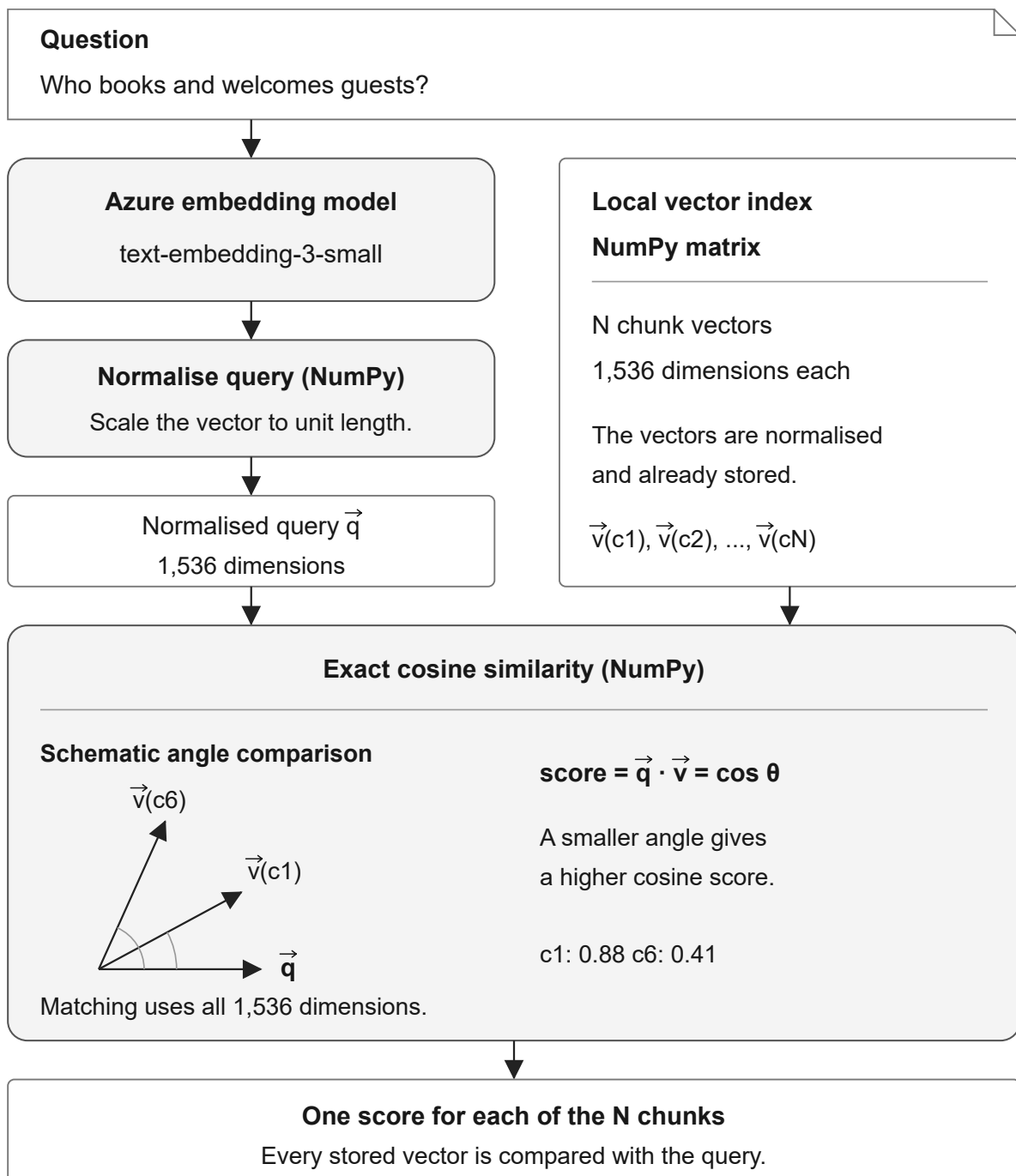


Figure 19. All stored vectors are compared with the query. The two-dimensional angle view is schematic; matching uses all 1,536 dimensions. Scores are constructed.

Thesis: Section 2.5; Appendix I.

Rank and select five full chunks

Retrieval

Rank every chunk by decreasing cosine similarity. Exact score ties retain the stored index order.

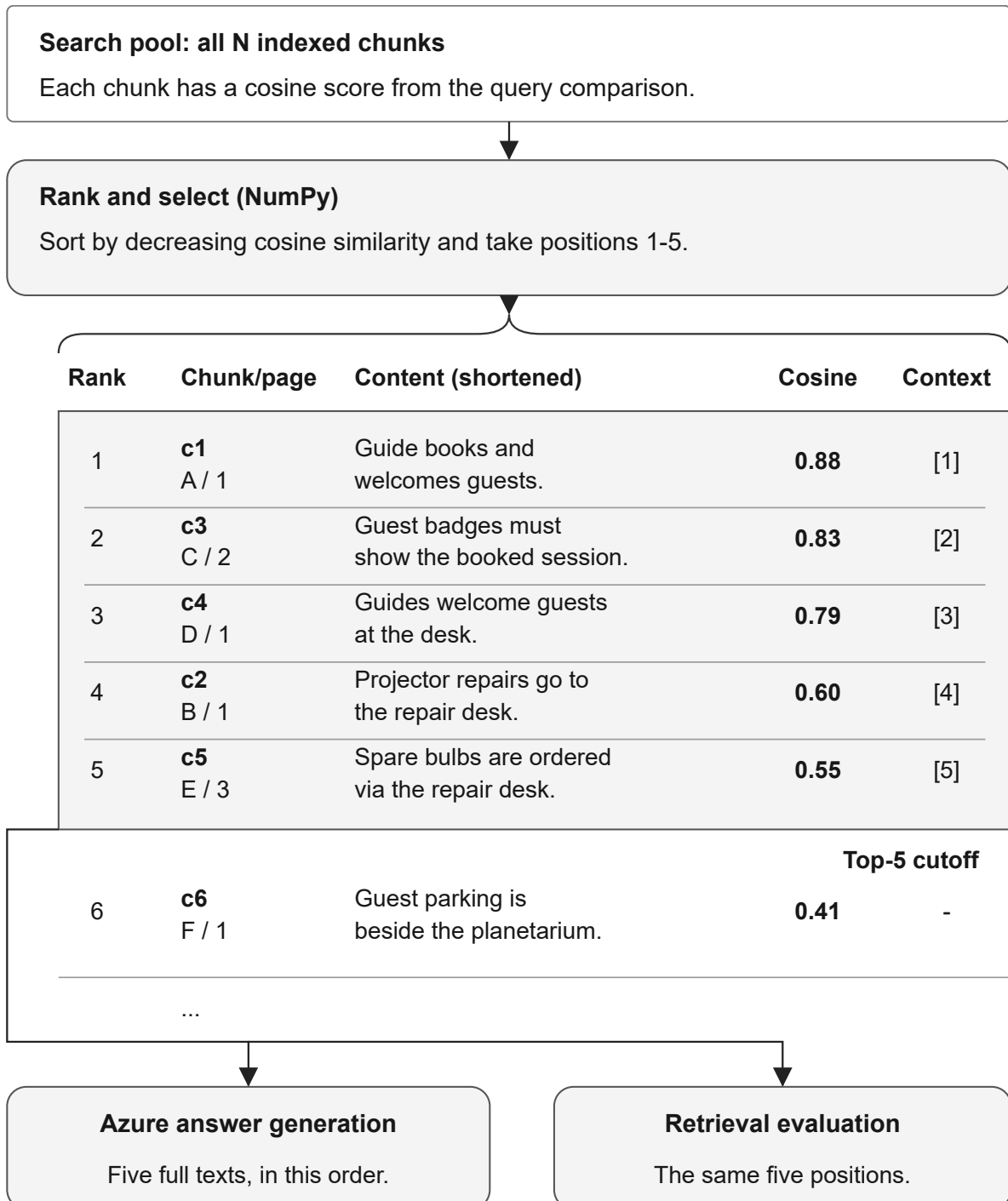


Figure 20. The table shows six constructed positions within the complete search pool. The first five full texts feed answer generation in this order. Retrieval evaluation uses the same five positions. Displayed excerpts are shortened.

Thesis: Section 2.5; Appendix I.

Generate an answer and trace its citation

Generation

Azure OpenAI generates the answer from the question and five complete chunks. The prompt requires the predefined abstention response when evidence is insufficient.

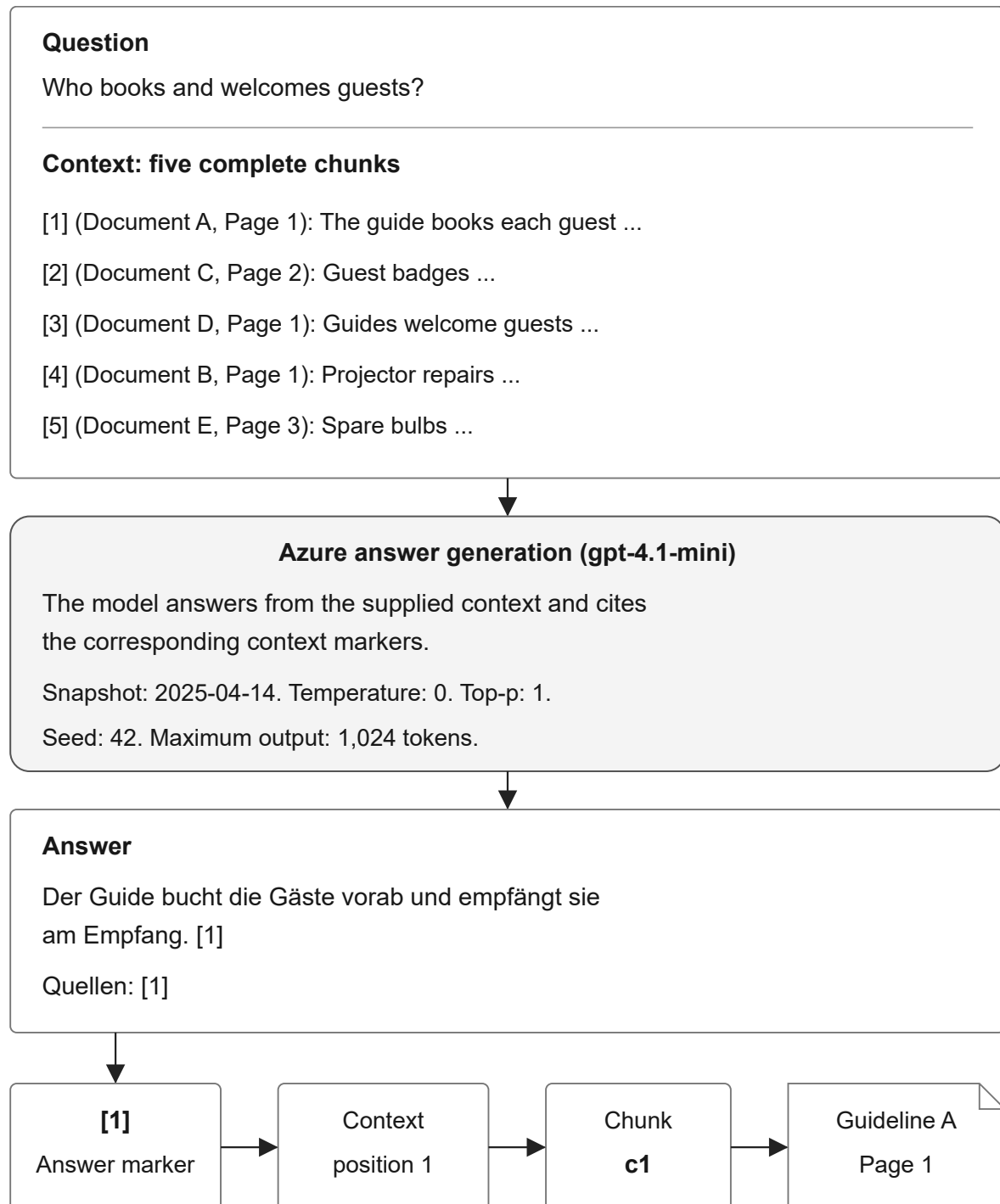


Figure 21. The German answer and Quellen marker illustrate the native C0 answer contract. Context excerpts are shortened only for display. A citation locates a source page; it does not establish factual support.

Thesis: Section 2.5; Appendix I.

Evaluate the retrieved ranking

Evaluation

Use the retained document-page mappings to attach shared qrel grades. Qrels evaluate the retrieval order; they do not determine it.

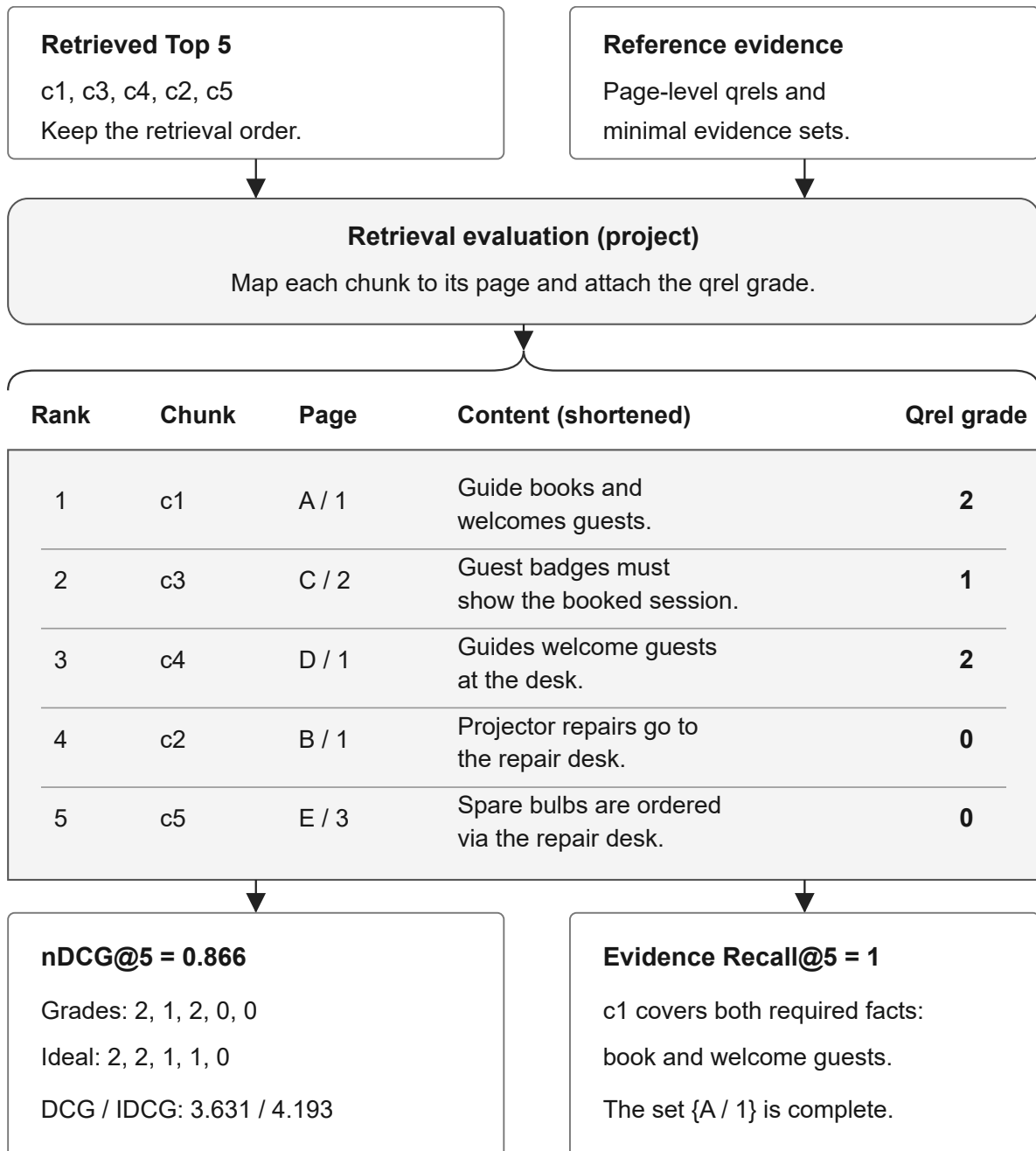


Figure 22. All five positions are evaluated, independently of answer citations. The judged pool contains six pages; F / 1 has grade 1. Linear gains give $nDCG@5 = 0.866$ [21]. The minimal evidence set {A / 1} is covered, so $Evidence\ Recall@5 = 1$.

Thesis: Sections 2.4, 2.6 and 2.7; Appendix I.

Scope and references

Reading the figures

These figures explain the baseline workflow. Controlled variants and their configuration are documented in the repository notebooks and methods pages.

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Notation: folded sheets and bordered tables show documents or retained records. Rounded rectangles show processing steps. Arrows show data flow. Vector components and two-dimensional geometry are schematic.

Shared evaluation: qrel grade 2 indicates direct support for a required fact, grade 1 related content and grade 0 no answer evidence. Retrieval metrics and source traceability are separate from semantic answer assessment.

Method details: docs/classical_rag_artifacts.md, docs/graphrag_artifacts.md and docs/evaluation_methods.md. The runnable public example uses RFC 2795; it is separate from these schematic figures.

Figure and reference numbers match the thesis appendices. The full bibliography for the citations used here follows.

Master's thesis

JH. From Guidelines to Grounded Answers: Answer Quality and Configuration Effects in Vector RAG and Microsoft GraphRAG Local Search. Master's thesis, University of Applied Sciences Technikum Wien, 2026.

[17] OpenAI. Vector embeddings. API documentation. <https://developers.openai.com/api/docs/guides/embeddings>

[21] K. Järvelin and J. Kekäläinen. Cumulated gain-based evaluation of IR techniques. ACM Transactions on Information Systems 20(4), 422-446, 2002. <https://doi.org/10.1145/582415.582418>

[38] C. D. Manning, P. Raghavan and H. Schütze. Introduction to Information Retrieval. Cambridge University Press, 2008, Sections 6.3 and 8.4. <https://nlp.stanford.edu/IR-book/>